



Introduction to Calculus selective

Part A: Limits and Continuity

Unit

TEACHER'S COPY

Q 1

Barker

Find: $\lim_{x \rightarrow 0} \frac{x^2 - 4x}{2x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{x(x-4)}{2x} &= \lim_{x \rightarrow 0} \frac{x-4}{2} \\ &= \frac{0-4}{2} = -2 \end{aligned}$$

Q 2

Barker

Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

(A) 2

(B) 1

(C) undefined

(D) 0

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} &= \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} \\ &= \lim_{x \rightarrow 1} x+1 \\ &= 1+1 = 2 \end{aligned}$$

Q 3

Barker

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}$

$$\begin{aligned} &= \lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)} \\ &= \lim_{x \rightarrow 3} (x+2) \\ &= 5 \end{aligned}$$

Q 4

James
Ruse

Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$.

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)} \\ &= \lim_{x \rightarrow 2} (x^2 + 2x + 4) \\ &= 12 \end{aligned}$$

Q 5

SYDNEY
BOYS

Find: $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$

$$\begin{aligned} &\lim_{x \rightarrow 3} \frac{(x-3)(x^2 + 3x + 9)}{x-3} \\ &= \lim_{x \rightarrow 3} x^2 + 3x + 9 \\ &= 3^2 + 3(3) + 9 \\ &= 27 \end{aligned}$$





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Part B: First Principles

2 unit

TEACHER'S COPY

Q1

Which of the following is the correct expression for differentiating $f(x) = x^2 - 2x$ from first principles?

ACE

(A) $f'(x) = \lim_{h \rightarrow 0} \frac{(x-h)^2 - 2(x-h) - (x^2 - 2x)}{h}$

(B) $f'(x) = \lim_{h \rightarrow 0} \frac{(x-h)^2 - 2(x-h) - (x^2 - 2x)}{h}$

(C) $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) + (x^2 - 2x)}{h}$

(D) $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - (x^2 - 2x)}{h}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - (x^2 - 2x)}{h}$$

Q2

Differentiate $f(x) = x^2 + 3$ from first principles.

4

CRANBROOK

ie: using $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$f(x+h) = (x+h)^2 + 3$$

$$= x^2 + 2xh + h^2 + 3$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 + 3) - (x^2 + 3)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3 - x^2 - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$$

Cancel h's

$$= \lim_{h \rightarrow 0} 2x + h$$

Let $h \rightarrow 0$

$$= 2x$$

Check answer to sure

$$\frac{d}{dx} x^2 + 3$$

$$= 2x$$

Q3

Differentiate by first principles $y = x^2 + 4x - 6$.

4

c) $y = x^2 + 4x - 6$

$f(x) = x^2 + 4x - 6$

$f(x+h) = (x+h)^2 + 4(x+h) - 6$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 4(x+h) - 6 - (x^2 + 4x - 6)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 4x + 4h - 6 - x^2 - 4x + 6}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 4h}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h + 4$$

$$= 2x + 4$$



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Part B: First Principles

Unit

TEACHER'S COPY

Q 4

CRANBROOK

Use the definition of the derivative, $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, to find $f'(x)$

when $f(x) = \frac{1}{x}$.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \text{ where } f(x) = \frac{1}{x}. \\ &= \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{x(x+h)} \times \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} \\ &= \frac{-1}{x(x+0)} = \frac{-1}{x^2} \end{aligned}$$

Q 5

JAMES RUSE

Find the derivative of $f(x) = 2x^2 - 9x$ from first principles.

$$\begin{aligned} 4. \quad f(x) &= 2x^2 - 9x \\ f(x+h) &= 2(x+h)^2 - 9(x+h) \\ &= 2(x^2 + 2xh + h^2) - 9x - 9h \\ &= 2x^2 + 4xh + 2h^2 - 9x - 9h \\ f(x+h) - f(x) &= 2x^2 + 4xh + 2h^2 - 9x - 9h - (2x^2 - 9x) \\ &= 4xh + 2h^2 - 9h \\ f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 9h}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(4x + 2h - 9)}{h} \\ &= \lim_{h \rightarrow 0} 4x + 2h - 9 \\ &= 4x - 9 \end{aligned}$$



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Part C: Derivatives

Unit

TEACHER'S COPY

Q 1

Differentiate with respect to x . 1

$$2x^4 + 7x^2 + 3$$

ACE

$$\frac{d}{dx}(2x^4 + 7x^2 + 3) = 8x^3 + 14x$$

Q 2

Differentiate with respect to x . $7x^5 - 4x^2 + 6$

ACE

$$\frac{d}{dx}(7x^5 - 4x^2 + 6) = 35x^4 - 8x$$

Q 3

Differentiate with respect to x . $3x^4 + 7x^2 + 1$

ACE

$$\frac{d}{dx}(3x^4 + 7x^2 + 1) = 12x^3 + 14x$$

Q 4

Differentiate: $\frac{3}{2}x^2 + 7$

ASCHAM

$$y' = 3x$$

Q 5

Differentiate: $\frac{2}{3x^2}$

ASCHAM

$$y' = -\frac{4}{3x^3} = -\frac{4}{x^3}$$

(2)

Q 6

Find the derivative of $f(x) = 2x^3 - \frac{x}{4} + 7$

ASCHAM

$$f(x) = 2x^3 - \frac{x}{4} + 7$$

$$f'(x) = 6x^2 - \frac{1}{4}$$

1

Q 7

Given that $x = 2t^4 + 100t^3$, find $\frac{dx}{dt}$ and hence the values of t for which $\frac{dx}{dt} = 0$

ASCHAM

$$x = 2t^4 + 100t^3$$

$$\frac{dx}{dt} = 8t^3 + 300t^2$$

$$0 = 8t^3 + 300t^2$$

$$= 4t^2(2t + 75)$$

$$\therefore t = 0 \quad \text{OR} \quad t = -37\frac{1}{2}$$

2





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Part C: Derivatives

Unit

TEACHER'S COPY

Q 8

Differentiate

$$\frac{1}{2}x^3 - x$$

(1)

ASCHAM

$$\frac{d}{dx} \left(\frac{1}{2}x^3 - x \right) = \frac{3}{2}x^2 - 1$$

Q 9

Differentiate

$$\frac{x^2 + 2x}{\sqrt{x}}$$

(2)

ASCHAM

$$\begin{aligned} \frac{d}{dx} \frac{x^2 + 2x}{\sqrt{x}} &= \frac{d}{dx} \left(x^{1/2} + 2x^{1/2} \right) \\ &= \frac{3}{2}x^{-1/2} + x^{-1/2} \checkmark \\ &= \frac{3\sqrt{x}}{2} + \frac{1}{\sqrt{x}} \checkmark \quad (2) \end{aligned}$$

Q10

Differentiate

$$\frac{1}{3x^2}$$

(2)

ASCHAM

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{3x^2} \right) &= \frac{d}{dx} \frac{1}{3} x^{-2} \\ &= -\frac{2}{3} x^{-3} \\ &= -\frac{2}{3x^3} \checkmark \quad (2) \end{aligned}$$

Q11

Differentiate

$$\frac{5x^3 - x}{x^2}$$

BAULKHAM

$$\begin{aligned} y &= \frac{5x^3 - x}{x^2} = 5x - \frac{1}{x} \\ y' &= 5 + \frac{1}{x^2} \checkmark \end{aligned}$$

Q12

Differentiate

$$x^3 - 2x^2 + 1$$

BAULKHAM

$$\begin{aligned} y &= x^3 - 2x^2 + 1 \\ \frac{dy}{dx} &= 3x^2 - 4x + 0 \end{aligned}$$



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Part C: Derivatives

Unit

TEACHER'S COPY

Q13

Differentiate $y = 6x^2 - 8x + 7$

1

BAULKHAM

$$\frac{dy}{dx} = 12x - 8 \quad \checkmark$$

Q14

Differentiate $y = 2\sqrt{x} - \frac{4}{x}$

2

BAULKHAM

$$y = 2x^{\frac{1}{2}} - 4x^{-1}$$

$$\frac{dy}{dx} = 1x^{-\frac{1}{2}} + 4x^{-2}$$

$$\checkmark \quad \checkmark$$

Q14

Differentiate $y = 5x^3 - \frac{1}{3x^2}$

2

SCEGGS

$$i) y = 5x^3 - \frac{1}{3}x^{-2} \quad \checkmark$$

$$\frac{dy}{dx} = 15x^2 + \frac{2}{3}x^{-3} \quad \checkmark$$

$$= 15x^2 + \frac{2}{3x^3}$$

Q15

Differentiate $f(x) = \frac{1}{x^3}$ (2)

BRIGIDINE

$$f(x) = \frac{1}{x^3}$$

$$f(x) = x^{-3} \quad \checkmark$$

$$f'(x) = -3x^{-4} \quad \checkmark$$

Q16

Differentiate $g(x) = \frac{5x}{2\sqrt{x}}$

BRIGIDINE

$$ii) g(x) = \frac{5x}{2\sqrt{x}}$$

$$g(x) = \frac{5x^{\frac{1}{2}}}{2x^{\frac{1}{2}}} \quad \checkmark$$

$$g'(x) = \frac{5}{2} \times \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{5}{4} x^{-\frac{1}{2}}$$

$$= \frac{5}{4\sqrt{x}} \quad \checkmark$$

Q17

If $f(x) = \frac{(2x+1)(2x+1)}{x}$, find $f'(x)$ (3)

BRIGIDINE

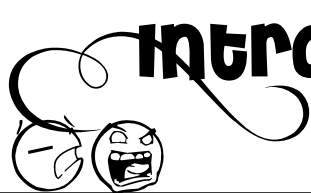
$$f(x) = \frac{(2x+1)(2x+1)}{x}$$

$$f(x) = \frac{4x^2 + 4x + 1}{x} \quad \checkmark$$

$$f(x) = \frac{4x^2}{x} + \frac{4x}{x} + \frac{1}{x}$$

$$f(x) = 4x + 4 + x^{-1} \quad \checkmark$$

$$f'(x) = 4 - x^{-2} \quad \checkmark$$



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Part C: Derivatives

Unit

TEACHER'S COPY

Q18

Differentiate with respect to x : $5x^2 - 5x + 3$

1

CRANBROOK

$$\frac{d}{dx}(5x^2 - 5x + 3)$$

$$= 10x - 5 \quad \checkmark$$

Q19

Differentiate with respect to x : $\frac{x^2 + 3x - 5}{x}$

2

CRANBROOK

$$\text{iii) } \frac{d}{dx} \left(\frac{x^2 + 3x - 5}{x} \right) = \frac{d}{dx} (x + 3 - 5x^{-1}) \quad \checkmark$$

$$= 1 + 5x^{-2} \quad \checkmark$$

$$= 1 + \frac{5}{x^2} \quad \checkmark$$

Only one term on denominator
Divide through!!
Not Quotient rule

Q20

Differentiate with respect to x : $\frac{1}{x^4}$

2

CRANBROOK

$$\text{iv) } \frac{d}{dx} \left(\frac{1}{x^4} \right) = -4x^{-5} \quad \checkmark$$

$$= \frac{-4}{x^5} \quad \checkmark$$

Q21

For the curve $y = ax^2 + bx + c$ where a, b, c are constants, it is given that when $x = 1, y = 1$ and $\frac{dy}{dx} = 1$. Show that $a = c$.

3

GIRAWEEEN

$$\text{(c) } y = ax^2 + bx + c \quad \text{--- (1)}$$

$$\frac{dy}{dx} = 2ax + b \quad \text{--- (2)}$$

$$\text{Sub } x=1, y=1 \text{ in (1)}$$

$$1 = a + b + c$$

$$\text{Sub } x=1, \frac{dy}{dx} = 1 \text{ in (2)}$$

$$1 = 2a + b$$

$$\text{By subtraction (3)}$$

$$a - c = 0$$

$$a = c$$

Q22

Find the derivative of: $5x^3 + 4x - 7 + \frac{3}{x^2}$

2

BRIGIDINE

$$y = 5x^3 + 4x - 7 + 3x^{-2}$$

$$\frac{dy}{dx} = 15x^2 + 4 - \frac{6}{x^3} \quad \text{(2)}$$



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Q23 If $S(t) = 5t^3 - 4t^2 + 5t$, find the value of $S'(-2)$

GIRAWEEEN

$$\begin{aligned} S(t) &= 5t^3 - 4t^2 + 5t \\ S'(t) &= 15t^2 - 8t + 5 \\ S'(-2) &= 15(-2)^2 - 8(-2) + 5 \\ &= 81 \end{aligned} \quad (2)$$

Q24 Differentiate: $y = 3\sqrt[3]{x} + 5x - 4$

GIRAWEEEN

$$\begin{aligned} (a) (i) \quad y &= 3x^{\frac{1}{3}} + 5x - 4 \\ y' &= 3 \times \frac{1}{3} x^{\frac{1}{3}-1} + 5 \\ &= x^{-\frac{2}{3}} + 5 = \underline{\underline{\sqrt[3]{x^{-2}} + 5}} \end{aligned} \quad (2)$$

Q25 Differentiate $y = \frac{1}{2}(x^4 - 4x^3 - 1)$

INDEPENDENT

$$\begin{aligned} y' &= \frac{1}{2}(4x^3 - 12x^2) \\ &= \underline{\underline{2x^3 - 6x^2}} \end{aligned}$$

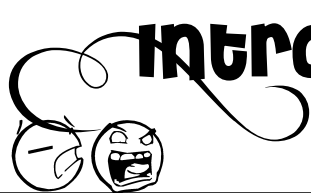
2

Q26 Given $f(x) = (x+1)\sqrt{x}$ find $f'(x)$.

NSBHS

$$\begin{aligned} f(x) &= (x+1)\sqrt{x} \\ &= x\sqrt{x} + \sqrt{x} \\ &= x^{\frac{3}{2}} + x^{\frac{1}{2}} \\ f'(x) &= \frac{d}{dx} x^{\frac{3}{2}} + \frac{d}{dx} x^{\frac{1}{2}} \\ &= \frac{3}{2} x^{\frac{1}{2}} + \frac{1}{2} x^{-\frac{1}{2}} \\ &= \frac{3\sqrt{x}}{2} + \frac{1}{2\sqrt{x}} \\ &= \frac{3(\sqrt{x})^2}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} \\ &= \frac{3x+1}{2\sqrt{x}} \\ &= \frac{3x+1}{2\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} \\ &= \frac{\sqrt{x}(3x+1)}{2x} \end{aligned}$$

2



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Part C: Derivatives

Unit

TEACHER'S COPY

Q27

Differentiate $f(x) = \frac{(x^2+3)(5x-1)}{x^2}$ 3

NSGHS

$$f(x) = \frac{5x^3 - x^2 + 15x - 3}{x^2}$$

$$= 5x - 1 + 15x^{-1} - 3x^{-2}$$

$$f'(x) = 5 - 15x^{-2} + 6x^{-3}$$

$$= 5 - \frac{15}{x^2} + \frac{6}{x^3}$$

Q28

Differentiate with respect to x: $\frac{x^3-4x^2}{\sqrt{x}}$ 2

NSGHS

$$y = \frac{x^3-4x^2}{\sqrt{x}} = x^{5/2} - 4x^{3/2}$$

$$\frac{dy}{dx} = \frac{5}{2}x^{3/2} - 6x^{1/2}$$

$$= \frac{5\sqrt{x^3} - 6\sqrt{x}}{2}$$

Q29

Differentiate with respect to x: $\left(x^2 + \frac{1}{x}\right)^2$ 2

NSGHS

$$iv) y = \left(x^2 + \frac{1}{x}\right)^2 = x^4 + 2x + x^{-2}$$

$$\frac{dy}{dx} = 4x^3 - 2x^{-3} + 2$$

$$= 4x^3 - \frac{2}{x^3} + 2$$

Q30

Given the function $f(x) = x^2 - 3x - 3$, find the values of b for which $f'(b) = f(b)$. 2

NSGHS

$$f(x) = x^2 - 3x - 3$$

$$f'(x) = 2x - 3$$

$$\therefore f(b) = b^2 - 3b - 3$$

$$f'(b) = 2b - 3$$

$$f'(b) = f(b)$$

$$\therefore 2b - 3 = b^2 - 3b - 3$$

$$b^2 - 5b = 0$$

$$b(b - 5) = 0$$

$$\therefore b = 0, 5$$



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Part C: Derivatives

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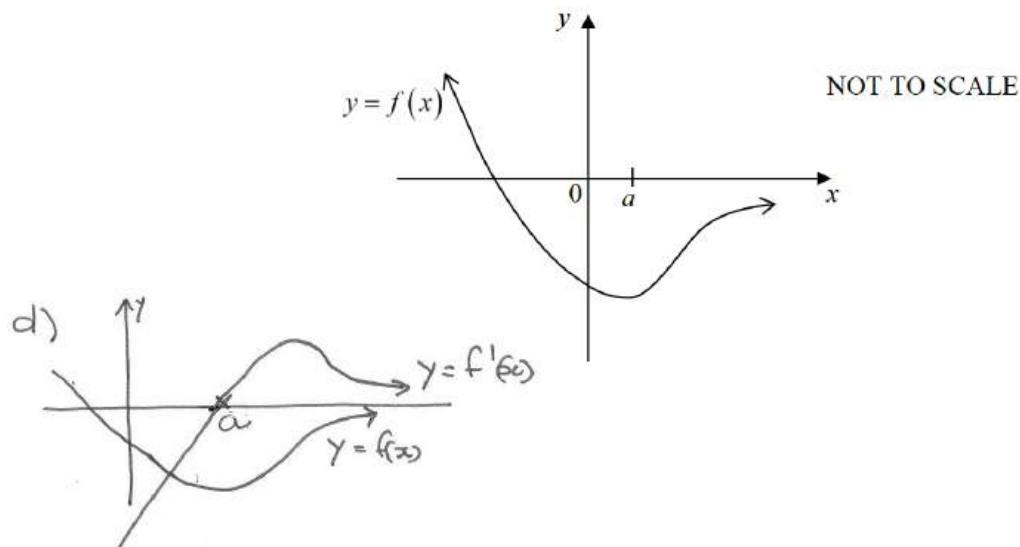
TEACHER'S COPY

Q31

Copy the following graph of $y = f(x)$ into your answer booklet.

Barker

Hence, clearly sketch and label $y = f'(x)$



Q32

Given that $f(x) = x^2 + x$, find the values of b for which $f''(b) = f(b)$

Barker

$$\begin{aligned} f(x) &= x^2 + x \\ f'(x) &= 2x \\ f''(x) &= 2 \\ f(b) &= b^2 + b \\ f''(b) &= 2 \\ 2 &= b^2 + b \\ 0 &= b^2 + b - 2 \\ 0 &= (b+2)(b-1) \\ \therefore b &= -2, 1 \end{aligned}$$

Q33

Differentiate the following functions with respect to x : $(3x^2 + 4)^7$

James
Ruse

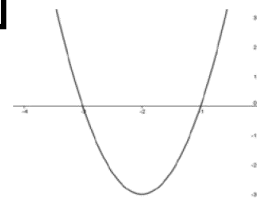
$$\begin{aligned} y &= (3x^2 + 4)^7 \\ y' &= 7(3x^2 + 4)^6 (6x) \\ &= 42x(3x^2 + 4)^6 \end{aligned}$$

Q34

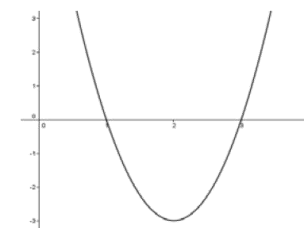
The graph of the gradient function of $y = x^3 + 6x^2 + 9x$ is:

James
Ruse

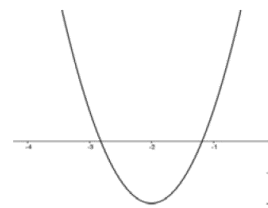
A.



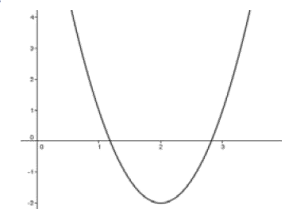
B.



C.



D.



$$\begin{aligned} \frac{dy}{dx} &= 3x^2 + 12x + 9 \\ &= 3(x^2 + 4x + 3) \\ &= 3(x+3)(x+1) \end{aligned}$$



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Part C: Derivatives

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Q35

Given that $y = (3x + 1)^2$. Show that $y - y' + 4 = x(y' - 12) - y$.

James
Ruse

c) $y = (3x + 1)^2$

show $y - y' + 4 = x(y' - 12) - y$ 30

$y' = 2(3x + 1) \cdot 3$

$= 6(3x + 1) \text{ or } 18x + 6$

RHS

$x(18x + 6 - 12) - (3x + 1)^2$

$= 18x^2 - 6x - (9x^2 + 6x + 1)$

$= 18x^2 - 6x - 9x^2 - 6x - 1$

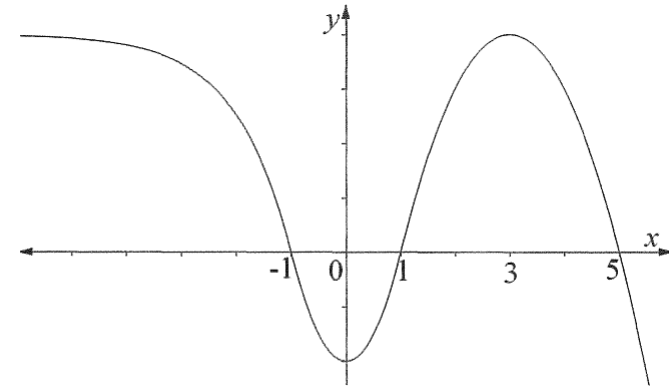
$= 9x^2 - 12x - 1$

LHS
 $(3x + 1)^2 - (18x + 6) + 4$
 $= 9x^2 + 6x + 1 - 18x - 6 + 4$
 $= 9x^2 - 12x - 1$

LHS = RHS

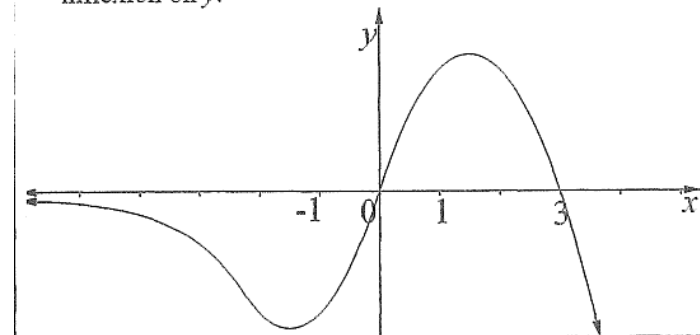
Q36

HURLSTONE



Sketch the derivative function of the curve above.

- (f) At stationary points derivative = 0. $\therefore y' = 0$ at $x = 0, 3$.
Curve decreasing when $x < 0$ and $x > 3$, and increasing when $0 < x < 3$.
 $\therefore y'$ is negative when $x < 0$ and $x > 3$, and y' positive when $0 < x < 3$.
Maximum, minimum values of y' occur at points of inflexion on y .



Q 1

Find the equation of the normal to the curve $y = 3x^2 - 1$ at the point (2,11)

3

ACE

$$y = 3x^2 - 1, \quad \frac{dy}{dx} = 6x$$

$$\text{At } (2,11) \quad \frac{dy}{dx} = 6 \times 2 = 12$$

Normal is perpendicular to the gradient of the tangent

$$m_1 m_2 = -1, \quad m_1 \times 12 = -1, \quad m_1 = -\frac{1}{12}$$

Equation of the normal at (2,11)

$$y - y_1 = m(x - x_1)$$

$$y - 11 = -\frac{1}{12}(x - 2)$$

$$12y - 132 = -x + 2$$

$$x + 12y - 134 = 0$$

Q2

Find the equation of the normal to the curve $f(x) = x^2 - 4x + 1$ at the point where $x = 1$.

2

ACE

$$f(x) = x^2 - 4x + 1$$

$$\text{At } x = 1 \quad f(1) = 1^2 - 4 \times 1 + 1 = -2$$

$$f'(x) = 2x - 4$$

$$\text{At } x = 1 \quad f'(1) = 2 \times 1 - 4 = -2$$

Normal is perpendicular to the gradient of the tangent

$$m_1 m_2 = -1, \quad m_1 \times -2 = -1, \quad m_1 = \frac{1}{2}$$

Equation of the normal at (1, -2)

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = \frac{1}{2}(x - 1)$$

$$2y + 4 = x - 1$$

$$x - 2y - 5 = 0$$

Q3

Find the equation of the tangent to the curve $y = \sqrt{x+1}$ at the point (3,2).

3

AMP

$$\frac{dy}{dx} = \frac{1}{2}(x+1)^{-\frac{1}{2}} \checkmark$$

$$\text{at } (3,2) \quad \frac{dy}{dx} = \frac{1}{2}(3+1)^{-\frac{1}{2}} = \frac{1}{4} \checkmark$$

equation of tangent

$$y - 2 = \frac{1}{4}(x - 3)$$

$$4y - 8 = x - 3$$

$$x - 4y + 5 = 0 \checkmark$$

Q4

Find, in general form, the equation of the normal to the curve $y = x^2 + 3x$ at the point (1,4) on it. (5)

ASCHAM

$$\begin{aligned} y' &= 2x + 3 \\ \text{When } x &= 1 \quad \text{grad of tan} &= 5 \\ \therefore \text{grad of normal} &= -\frac{1}{5} \\ \text{eqn: } y - 4 &= -\frac{1}{5}(x - 1) \\ 5y - 20 &= -x + 1 \\ x + 5y - 21 &= 0 \end{aligned}$$

Q5

At which points are the tangents to the curve $y = x^3 - 6x^2 + 2x - 9$ parallel to the line $y = -7x + 4$?

3

ASCHAM

c1 $y = -7x + 4$ has gradient of -7
 for $y = x^3 - 6x^2 + 2x - 9$
 $\frac{dy}{dx} = 3x^2 - 12x + 2$
 $\therefore 3x^2 - 12x + 2 = -7$ $\therefore$ parallel.
 $3x^2 - 12x + 9 = 0$
 $3(x^2 - 4x + 3) = 0$
 $3(x-3)(x-1) = 0$
 $\therefore x = 3$ OR $x = 1$
 for $x = 3$: $y = 3^3 - 6(9) + 6 - 9$
 $= -30$
 $\therefore (3, -30)$
 for $x = 1$: $y = 1^3 - 6(1) + 2 - 9$
 $= -12$
 $\therefore (1, -12)$

Q6

Find the equation of the normal to the curve $y = x^2 + \frac{5}{x} - 2$ at the point where $x = 1$.

(3)

ASCHAM

b) $y = x^2 + \frac{5}{x} - 2$
 $y' = 2x - \frac{5}{x^2}$
 $x = 1$ $y' = 2 - 5 = -3$ $y = 4$
 grad of tangent $= -3$
 grad of normal $= \frac{1}{3}$
 $y - 4 = \frac{1}{3}(x - 1)$
 $3y - 12 = x - 1$
 $x - 3y + 11 = 0$ ✓

Introduction to Calculus selective

Part D: Tangents and Normals

2 unit

TEACHER'S COPY

Q7

Find the equation of the tangent to the curve $y = x^3 - 3x^2 + 3x - 1$ at the point where $x = 2$

BAULKHAM

$$\begin{aligned} \text{At } x = 2, y &= 8 - 12 + 6 - 1 \\ &= 1 \\ y' &= 3x^2 - 6x + 3 \\ \text{at } x = 2, y' &= 12 - 12 + 3 = 3 \\ \therefore \text{tangent is} \\ y - 1 &= 3(x - 2) \\ y &= 3x - 5 \end{aligned}$$

Q8

Find the equation of the normal to the curve $y = \sqrt{x+2}$ at the point where $x = 7$

BAULKHAM

$$\begin{aligned} x &= 7, y = 3 \\ m_T &= \frac{1}{2}(x+2)^{-\frac{1}{2}} \\ x = 7, m_T &= \frac{1}{2\sqrt{9}} = \frac{1}{6} \\ \therefore m_n &= -6 \\ y - 3 &= -6(x - 7), \quad 6x + y - 45 = 0 \end{aligned}$$

3

Q9

Find the value(s) of k if the normal to the curve $y = kx^2 - x - 3$ is parallel to the line $5x - 3y + 2 = 0$ at the point where $x = 1$. (3)

BAULKHAM

$$\begin{aligned} y &= kx^2 - x - 3 \\ \frac{dy}{dx} &= 2kx - 1 \\ \text{when } x &= 1 \\ m \text{ of tang} &= 2k - 1 \\ \text{gradient of} & \\ \text{tangent} &= 2k - 1 \\ m \text{ of normal} &= -\frac{1}{2k-1} \quad \text{or mark for: } 5x - 3y + 2 = 0 \\ \text{or} & \\ 3y &= 5x + 2 \\ y &= \frac{5}{3}x + \frac{2}{3} \\ \therefore m &= \frac{5}{3} \\ \text{Since normal is parallel then} \\ -\frac{1}{2k-1} &= \frac{5}{3} \\ -3 &= 5(2k-1) \\ -3 &= 10k-5 \\ 2 &= 10k \\ k &= \frac{1}{5} \end{aligned}$$

Q10

Find the equation of the normal to the curve $x^2 = 8y$ at the point where $x = 2$

4

BAULKHAM

$x^2 = 8y$
 $\Rightarrow y = \frac{x^2}{8}$ *Make y subject*
 $\frac{dy}{dx} = \frac{x}{4}$ *must be able to differentiate*
 let $x = 2$
 $m = \frac{1}{2}$ *tangent*
 Normal: $m_n = -2$ *Generally poorly done.*
 $y - y_1 = m(x - x_1)$
 Need y_1
 When $x = 2$
 $y = \frac{4}{8}$ *(sub into $y = \frac{x^2}{8}$)*
 $y = \frac{1}{2}$
 $y - \frac{1}{2} = -2(x - 2)$
 $y = -2x + 4\frac{1}{2}$ *lots of mistakes bringing 2 across*

Q11

Find the equation of the normal to the curve $y = 3 + 6x - 2x^2$ at the point (1, 7).

4

BAULKHAM

d) $y = 3 + 6x - 2x^2$
 $\frac{dy}{dx} = 6 - 4x$
 At $x = 1$
 $m_{\text{tangent}} = 2$
 $m_{\text{normal}} = -\frac{1}{2}$ *pt (1, 7)*
 Equation of normal:
 $y - y_1 = m(x - x_1)$
 $y - 7 = -\frac{1}{2}(x - 1)$
 $2y - 14 = -x + 1$ ④
 $x + 2y - 15 = 0$

Q12

Find the equation of the normal to the parabola $y = 2x^2 - 5x + 1$ at the point on the parabola where the gradient is 3.

4

GIRAWEN

(c) $y = 2x^2 - 5x + 1$
 $y' = 4x - 5$
 $4x - 5 = 3$
 $x = 2$
 When $x = 2$, $y = -1$
 Equation of normal
 $y + 1 = -\frac{1}{3}(x - 2)$ ④
 $3y + 3 = -x + 2$
 $x + 3y + 1 = 0$

Q13

Show that the graph of $y = x^2 + 4x - 12$ crosses the x axis at two points.
Find the equations of the normals at these points.

GIRAWREEN

$$d) \ y = x^2 + 4x - 12$$

$$y = 0 \quad x^2 + 4x - 12 = 0 \\ (x+6)(x-2) = 0$$

$$x = -6 \text{ or } x = 2$$

$$\frac{dy}{dx} = 2x + 4 \quad \text{When } x = -6 \quad \frac{dy}{dx} = -8 \\ \text{When } x = 2 \quad \frac{dy}{dx} = 8$$

At $(-6, 0)$ Gradient of normal $= \frac{1}{8}$

Equation of normal at $(-6, 0)$

$$y - 0 = \frac{1}{8}(x + 6)$$

$$y = \frac{x}{8} + \frac{6}{8} \text{ or } x - 8y + 6 = 0$$

At $(2, 0)$ Gradient of normal $= -\frac{1}{8}$

Equation of normal at $(2, 0)$

$$y - 0 = -\frac{1}{8}(x - 2)$$

$$y = -\frac{x}{8} + \frac{1}{4} \text{ or } x + 8y - 2 = 0$$

Q14

Consider the curve $y = x + \frac{8}{x}$.

NSGHS

- | | | |
|------|---|---|
| i) | Find the gradient of the curve at the point P where $x = 2$. | 2 |
| ii) | What are the co-ordinates of P . | 1 |
| iii) | Find the equation of the normal to the curve at the point P . | 2 |

$$b) \ y = x + 8x^{-1}$$

$$\frac{dy}{dx} = 1 - 8x^{-2} \\ = 1 - \frac{8}{x^2}$$

$$x = 2, \quad \frac{dy}{dx} = 1 - \frac{8}{4} \\ = -1$$

The gradient at P is -1

$$(ii) \ y = 2 + \frac{8}{2} \\ = 6$$

P is $(2, 6)$

(iii) gradient of normal is 1

equation is

$$y - 6 = 1(x - 2)$$

$$y - 6 = x - 2$$

$$y = x + 4$$

Q15

At what point on the curve $f(x) = \frac{x^2}{4} - 5$ is the tangent parallel to $3x - y + 2 = 0$?

NSGHS

$$y = 3x + 2 \quad \text{gradient } 3$$

$$f(x) = \frac{1}{4}x^2 - 5$$

$$f'(x) = \frac{1}{2}x$$

$$f'(x) = 3$$

$$\frac{1}{2}x = 3$$

$$x = 6$$

$$x = 6, y = 4$$

at $(6, 4)$ the tangent is parallel to $3x - y + 2 = 0$

Q16

Find the equation of the normal to the curve $y = x^3 - 4x - 8$ at the point $(2, -8)$. Give your answer in general form.

NSGHS

$$y = x^3 - 4x - 8$$

$$\frac{dy}{dx} = 3x^2 - 4$$

$$m \text{ of tangent at } x=2 \text{ is } 3(2)^2 - 4 = 8$$

$$\therefore m \text{ of normal is } -\frac{1}{8}$$

$$y - y_1 = m(x - x_1)$$

$$y + 8 = -\frac{1}{8}(x - 2)$$

$$8y + 64 = -x + 2$$

$$x + 8y + 62 = 0$$

Q17

The gradient of the tangent to the curve $y = ax^2 - 8x + 10$ is 4 when $x = 2$. Find the value of a .

NSGHS

$$\frac{dy}{dx} = 2ax - 8$$

$$\text{when } x = 2,$$

$$4 = 2(2)a - 8$$

$$12 = 4a$$

$$\therefore a = 3$$

Q18

Show that the tangents at the points $(1, -3)$ and $(0, -3)$ on the curve $y = x^2 - x - 3$ are perpendicular to each other.

NSGHS

$$\frac{dy}{dx} = 2x - 1$$

So, gradient of tangent at $x = 1$ is 1.

When $x = 0,$

$$\frac{dy}{dx} = 2(0) - 1$$

$$= -1$$

So, gradient of tangent at $x = 0$ is -1 .

In perpendicular lines, $m_1 m_2 = -1$

$1 \times -1 = -1$, $\therefore$ tangents at $x = 1$ and $x = 0$ are perpendicular.

Q19

- (i) Find the equation of the tangent to $y = x^3 - 9x^2 + 20x - 8$ at the point $(1, 4)$. 3
(ii) For what values of x are the tangents parallel to the line $y = -4x + 3$? 2

NSGHS

b) $y = x^3 - 9x^2 + 20x - 8$ $(1, 4)$

i) $\frac{dy}{dx} = 3x^2 - 18x + 20$

then $x=1$, $m = 3(1)^2 - 18(1) + 20$
 $= 5$

$\therefore$ eqn tangent $y - 4 = 5(x - 1)$
 $y = 5x - 1$

ii) tangents parallel $\rightarrow m = -4$

$\therefore -4 = 3x^2 - 18x + 20$

$0 = 3x^2 - 18x + 24$

$= x^2 - 6x + 8$

$= (x - 4)(x - 2)$

$\therefore x = 2$ and $x = 4$.

Q20

- Find the equation of the tangent to the curve $y = \frac{x}{x+2}$ at the point where $x = -1$. Give your answer in general form. 4

SCEGGS

$y = \frac{x}{x+2}$

$\frac{dy}{dx} = \frac{(x+2) \times 1 - x \times 1}{(x+2)^2} \quad \left(\frac{vu' - uv'}{v^2} \right)$

$= \frac{2}{(x+2)^2} \quad \checkmark$

$x = -1 \rightarrow m_T = \frac{2}{(-1+2)^2} = 2 \quad \checkmark$

$y = \frac{-1}{-1+2} = -1 \quad \checkmark$

Equation of tangent:

$y - (-1) = 2(x - (-1))$

$y + 1 = 2x + 2$

$2x - y + 1 = 0 \quad \checkmark$

Q21

Barker

The curve $y = ax^3 + bx$ passes through the point $(1, 7)$. The tangent at this point is parallel to the line $y = 2x - 6$. Find the values of a and b .

$$\begin{aligned} b. \quad y &= ax^3 + bx \quad (1, 7) \\ 7 &= a(1)^3 + b(1) \\ 7 &= a + b \\ \\ y' &= 3ax^2 + b \\ y' &= 2 \text{ when } x = 1 \\ 2 &= 3a(1)^2 + b \\ 2 &= 3a + b \\ \\ \begin{cases} a + b = 7 \\ 3a + b = 2 \end{cases} \\ -2a &= 5 \\ a &= -\frac{5}{2} \\ \\ \therefore -\frac{5}{2} + b &= 7 \\ b &= \frac{19}{2} \\ \\ \therefore a &= -\frac{5}{2}, \quad b = \frac{19}{2} \end{aligned}$$

Q22

Barker

Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point $(3, 2)$

$$\begin{aligned} y &= x^2 - 3x + 2 \\ \frac{dy}{dx} &= 2x - 3 \\ x = 3 \quad \frac{dy}{dx} &= 2(3) - 3 \\ &= 3 = m \\ \text{then } m &= 3 \quad \text{At } (3, 2) \\ y - 2 &= 3(x - 3) \\ y - 2 &= 3x - 9 \\ y &= 3x - 7 \quad \text{or } 3x - y - 7 = 0 \end{aligned}$$

Q23

James
Ruse

- Show that the tangent to $y = x^3$ at the point where $x = 2$ is $y = 12x - 16$.
- Show that the tangent meets the curve again at $(-4, -64)$.

$$\begin{aligned} (i) \quad y &= x^3 \\ \frac{dy}{dx} &= 3x^2 \\ \text{when } x = 2 \quad y &= 8 \quad \text{then } \frac{dy}{dx} = 3(2^2) = 12 \\ \therefore \text{equation of tangent is} \\ y - 8 &= 12(x - 2) \\ y - 8 &= 12x - 24 \\ y - 12x + 16 &= 0 \\ \text{or} \\ y &= 12x - 16 \end{aligned}$$

$$\begin{aligned} (ii) \quad y &= x^3 \\ \text{Test } (-4, -64) \\ \text{LHS} &= -64 \\ \text{RHS} &= (-4)^3 = -64 \\ \text{or} \\ y &= 12x - 16 \\ \text{LHS} &= -64 \\ \text{RHS} &= 12(-4) - 16 = -48 - 16 = -64 \end{aligned}$$

Since LHS = RHS for both equations
we can say tangent and curve
meets again at $(-4, -64)$

Alternatively,

$$\begin{aligned} y &= x^3 \quad \text{and} \quad y = 12x - 16 \\ \therefore x^3 - 12x + 16 &= 0 \end{aligned}$$

Sub in $x = 4$ to get true statement
then sub in to find y in the two
equations



Introduction to Calculus selective

Part e: Chain Rule

Unit

TEACHER'S COPY

Q1

What is the derivative of $\sqrt{x^2+3}$?

ACE

(A) $\frac{1}{2\sqrt{x^2+3}}$

(B) $2\sqrt{x^2+3}$

(C) $x\sqrt{x^2+3}$

(D) $\frac{x}{\sqrt{x^2+3}}$

$$\begin{aligned}\frac{d}{dx}(\sqrt{x^2+3}) &= \frac{1}{2}(x^2+3)^{-\frac{1}{2}} \times 2x \\ &= \frac{x}{\sqrt{x^2+3}}\end{aligned}$$

Q2

Differentiate: $y = \sqrt[3]{4-x^2}$

ASCHAM

$$\begin{aligned}y' &= \frac{1}{3}(4-x^2)^{-\frac{2}{3}} \times -2x \\ &= \frac{-2x}{3(4-x^2)^{\frac{2}{3}}}\end{aligned}$$

(2)

Q3

Find the derivative of $y = \sqrt{5-x^2}$

ASCHAM

$$\begin{aligned}y &= \sqrt{5-x^2} \\ y' &= \frac{1}{2}(5-x^2)^{-\frac{1}{2}} \times -2x \\ &= -x(5-x^2)^{-\frac{1}{2}} \\ &= \frac{-x}{\sqrt{5-x^2}}\end{aligned}$$

2

Q4

Differentiate $\sqrt{x^2-6}$

BAULKHAM

$$\begin{aligned}y &= \sqrt{x^2-6} \\ y' &= \frac{1}{2}(x^2-6)^{-\frac{1}{2}} \times 2x \\ &= \frac{x}{\sqrt{x^2-6}}\end{aligned}$$

Q5

Differentiate $(2x+1)^8$

BAULKHAM

$$\begin{aligned}y &= (2x+1)^8 \\ y' &= 8(2x+1)^7 \times 2 \\ &= 16(2x+1)^7\end{aligned}$$



Introduction to Calculus selective

Part e: Chain Rule

Unit

TEACHER'S COPY

Q 6

Differentiate $y = (3x^3 - 7)^4$

2

BAULKHAM

$$\begin{aligned} y' &= 4(3x^3 - 7)^3 \times 9x^2 \quad \checkmark \\ &= 36x^2(3x^3 - 7)^3 \quad \checkmark \end{aligned}$$

Q 7

Differentiate with respect to x : $(2x+8)^5$

1

CRANBROOK

$$\begin{aligned} \frac{d}{dx} (2x+8)^5 \\ 5(2x+8)^4 \times 2 \quad \checkmark \\ \underline{10(2x+8)^4} \end{aligned}$$

Q 8

Find the derivative of: $(3x^2 - 7)^{10}$

2

GIRAWREEN

$$\begin{aligned} y &= (3x^2 - 7)^{10} \\ \frac{dy}{dx} &= 10x(3x^2 - 7)^9 \quad (2) \end{aligned}$$

Q 9

Find the derivative of: $\sqrt{4-x^2}$

2

GIRAWREEN

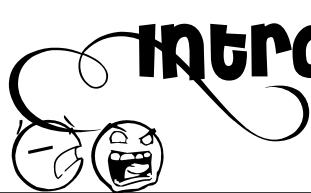
$$\begin{aligned} y &= (4-x^2)^{1/2} \\ y' &= \frac{1}{2}(4-x^2)^{-1/2} \cdot -2x \\ &= \frac{-x}{\sqrt{4-x^2}} \end{aligned}$$

Q 10

Differentiate: $y = \sqrt{x^2 - 2x}$

GIRAWREEN

$$\begin{aligned} (iv) \quad y &= \sqrt{x^2 - 2x} \\ y' &= \frac{1}{2\sqrt{x^2 - 2x}} \times 2x - 2 = \frac{2(x-1)}{2\sqrt{x^2 - 2x}} \quad (3) \\ &= \frac{x-1}{\sqrt{x^2 - 2x}} \end{aligned}$$



Introduction to Calculus selective

Part e: Chain Rule

Unit

TEACHER'S COPY

Q11

Differentiate with respect to x : $\sqrt{3x^2 - 2x}$ 2

BAULKHAM

$$\begin{aligned} &= \frac{d}{dx} (3x^2 - 2x)^{\frac{1}{2}} \\ &= \frac{1}{2} (3x^2 - 2x)^{-\frac{1}{2}} (6x - 2) \\ &= (3x - 1) (3x^2 - 2x)^{-\frac{1}{2}} \\ &= \frac{3x - 1}{\sqrt{3x^2 - 2x}} \end{aligned}$$

Q12

Find $g'(-1)$ for $g(x) = (-x^4 + 3)^5$. 2

NSBHS

$$\begin{aligned} g(x) &= (-x^4 + 3)^5 \\ \therefore g'(x) &= 5 \times (-x^4 + 3)^4 \times -4x^3 \\ &= -20x^3 (-x^4 + 3)^4 \\ \therefore g'(-1) &= -20(-1)^3 (-(-1)^4 + 3)^4 \\ &= -20(4)^4 \\ &= -20 \times 64 \\ &= -1280 \end{aligned}$$

Q13

Differentiate with respect to x , $y = (6x + 1)^{\frac{3}{2}}$ 2

NSGHS

$$\begin{aligned} y &= (6x + 1)^{\frac{3}{2}} \\ \frac{dy}{dx} &= \frac{3}{2} (6x + 1)^{\frac{1}{2}} \times 6 \\ &= \frac{18}{2} (6x + 1)^{\frac{1}{2}} \end{aligned}$$

Q14

Differentiate with respect to x : $\frac{1}{(2x + 3)^4}$ 2

GIRAWEEEN

$$\begin{aligned} y &= (2x + 3)^{-4} \\ \frac{dy}{dx} &= -4 (2x + 3)^{-5} \times 2 \\ &= \frac{-8}{(2x + 3)^5} \end{aligned}$$



Introduction to Calculus selective

Part e: Chain Rule

Unit

TEACHER'S COPY

Q15

Differentiate with respect to x :

$$(5-2x)^7$$

BARKER

$$\begin{aligned} \frac{d}{dx} (5-2x)^7 &= 7(5-2x)^6 \times -2 \\ &= -14(5-2x)^6 \end{aligned}$$

Q16

Differentiate the following with respect to x : $(2x^2 + 7)^3$

JAMES
RUSE

$$\begin{aligned} y &= (2x^2 + 7)^3 \\ \frac{dy}{dx} &= 3(2x^2 + 7)^2 \cdot 4x \\ &= 12x(2x^2 + 7)^2 \end{aligned}$$

Q17

Differentiate $(2x+1)^8$ with respect to x .

HURLSTONE

$$\begin{aligned} \frac{d[(2x+1)^8]}{dx} &= 8(2x+1)^7 \times 2 \\ &= 16(2x+1)^7 \end{aligned}$$

Q18

Differentiate $y = (x^4 - 1)^9$

HURLSTONE

$$\begin{aligned} \frac{dy}{dx} &= 9(4x^3)(x^4 - 1)^8 \\ &= 36x^3(x^4 - 1)^8 \end{aligned}$$

Q19

Which of the following is the derivative of x^{n+1} with respect to x ?

(A) nx^{n+1}

(B) $\frac{x^{n+1}}{n+1}$

(C) $(n+1)x^n$

(D) $(n+1)x^{n-1}$

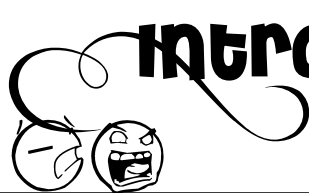
SYDNEY
GRAMMAR

Q20

Differentiate: $(3x-2)^3$

SYDNEY
GRAMMAR

$$\frac{d}{dx} (3x-2)^3 = 9(3x-2)^2 \checkmark$$



Introduction to Calculus selective

Part F: Product Rule

Unit

TEACHER'S COPY

Q1

Differentiate with respect to x . $(3x+2)^3(x+1)^4$

ACE

$$\begin{aligned}\frac{d}{dx}(3x+2)^3(x+1)^4 &= (3x+2)^3 \times 4(x+1)^3 + (x+1)^4 3(3x+2)^2 \times 3 \\ &= (3x+2)^2(x+1)^3[(3x+2)4 + (x+1)9] \\ &= (3x+2)^2(x+1)^3(21x+17)\end{aligned}$$

Q2

Differentiate the following $y = (x^3 + 5)^4$

AMP

$$\begin{aligned}\frac{dy}{dx} &= 4(x^3 + 5)^3 \times 3x^2 \checkmark \\ &= 12x^2(x^3 + 5)^3 \checkmark\end{aligned}$$

2

Q3

Differentiate: $x^2(2x-1)^5$

ASCHAM

$$\begin{aligned}y' &= x^2 \cdot 5(2x-1)^4 \times 2 + (2x-1)^5 \cdot 2x \\ &= 10x^2(2x-1)^4 + 2x(2x-1)^5 \\ &= 2x(2x-1)^4(5x + 2x-1) \\ &= 2x(2x-1)^4(7x-1)\end{aligned}$$

(2)

Q4

Find the derivative of $y = \frac{x(x-2)^6}{2}$

ASCHAM

$$\begin{aligned}y &= \frac{x(x-2)^6}{2} \\ \frac{dy}{dx} &= \frac{1}{2} \times [x \times 6(x-2)^5 + 1 \times (x-2)^6] \\ &= \frac{1}{2} [(x-2)^5(6x + (x-2))] \\ &= \frac{1}{2} [(x-2)^5(7x-2)]\end{aligned}$$

2



Introduction to Calculus selective

Part F: Product Rule

2 unit

TEACHER'S COPY

Q5

Differentiate

$$x^3(1-x)^4$$

ASCHAM

$$\begin{aligned}\frac{d}{dx} x^3(1-x)^4 &= x^3 \cdot 4(1-x)^3 \cdot (-1) + \\ &\quad (1-x)^4 \cdot 3x^2 \cdot 1 \quad \checkmark \\ &= x^2(1-x)^3(-4x + 3 - 3x) \quad \checkmark \\ &= x^2(1-x)^3(3-7x) \quad \checkmark\end{aligned}$$

Q6

Differentiate

$$y = (x^2 - 5)(2x + 1)$$

2

BAULKHAM

$$\begin{aligned}y' &= 2x(2x + 1) + (x^2 - 5)(2) \quad \checkmark \\ &= 6x^2 + 2x - 10 \quad \checkmark\end{aligned}$$

Q7

Differentiate $y = x(4x + 3)^5$

2

SCEGGS

$$\begin{aligned}y &= x(4x+3)^5 \\ \frac{dy}{dx} &= v \frac{dy}{dv} + u \frac{dv}{du} \\ &= (4x+3)^5 \times 1 + x \times 5(4x+3)^4 \times 4 \\ &= (4x+3)^4 [4x+3 + 20x] \\ &= (4x+3)^4 (24x+3)\end{aligned}$$

Q8

Differentiate with respect to x : $x^4(x^2 - 8)^5$

3

CRANBROOK

$$\begin{aligned}u v' + v u' \\ x^4 \times 5(x^2 - 8)^4 \times 2x + 4x^3(x^2 - 8)^5 \\ 10x^5(x^2 - 8)^4 + 4x^3(x^2 - 8)^5\end{aligned}$$

Q9

Find the derivative of: $3x^4(x - 4)^5$

3

GIRAWEEEN

$$\begin{aligned}y &= 3x^4(x-4)^5 \\ y' &= v u' + u v' \\ &= 12x^3(x-4)^5 + 15x^4(x-4)^4 \\ &= 3x^3(x-4)^4(4(x-4) + 5x) \\ &= 3x^3(x-4)^4(9x-16) \quad \textcircled{3}\end{aligned}$$



Introduction to Calculus selective

Part F: Product Rule

2 unit

TEACHER'S COPY

Q10

Differentiate: $y = (7x^3 - 1)(5x^2 - x)$

GIIRAWEEEN

$$(ii) y = (7x^3 - 1)(5x^2 - x)$$

$$y' = (7x^3 - 1)(10x - 1) + (5x^2 - x)21x^2$$

$$= 175x^4 - 28x^3 - 10x + 1 \quad (3)$$

Q11

Differentiate $y = x^3(x-1)^5$, leaving your answer in fully factorised form.

3

KNOX

$$(iv) y = x^3(x-1)^5$$

$$\frac{dy}{dx} = x^3 \cdot 5(x-1)^4 \cdot 1 + (x-1)^5 \cdot 3x^2$$

$$= x^2(x-1)^4(5x + 3x - 3)$$

$$= x^2(x-1)^4(8x - 3) \checkmark$$

Q12

Differentiate with respect to x , $y = x^2\sqrt{x+1}$

2

NSGHS

$$\frac{dy}{dx} = uv' + vu'$$

$$= x^2\left(\frac{1}{2}\right)(x+1)^{-\frac{1}{2}} + 2x(x+1)^{\frac{1}{2}}$$

$$= \frac{1}{2}x(x+1)^{-\frac{1}{2}}[x + 4(x+1)]$$

$$= \frac{1}{2}x(x+1)^{-\frac{1}{2}}(5x+1)$$

Q13

Differentiate with respect to x : $\frac{x}{2}(2x-5)^3$

2

NSGHS

$$= \frac{1}{2}(2x-5)^3 + \frac{x}{2} \times 6(2x-5)^2$$

$$= \frac{(2x-5)^3}{2} + \frac{6x(2x-5)^2}{2}$$

$$= \frac{(2x-5)^2(8x-5)}{2}$$



Introduction to Calculus selective

Part F: Product Rule

Unit

TEACHER'S COPY

Q14

Differentiate with respect to x : $x\sqrt{1+x}$

BARKER

$$\begin{aligned} y &= x(1+x)^{\frac{1}{2}} \\ \frac{dy}{dx} &= (1+x)^{\frac{1}{2}} \cdot 1 + x \cdot \frac{1}{2}(1+x)^{-\frac{1}{2}} \\ &= \sqrt{1+x} + \frac{x}{2\sqrt{1+x}} \end{aligned}$$

Q15

Differentiate with respect to x : $y = 4x^2(6x+2)^5$

SYDNEY
BOYS

$$\begin{aligned} y' &= 8x(6x+2)^5 + 120x^2(6x+2)^4 \\ &= 8x(6x+2)^4 \{6x+2 + 15x\} \\ &= 8x(6x+2)^4(21x+2) \end{aligned}$$

Q16

Differentiate with respect to x : $y = (x-1)(x^2+7x)^5$

SYDNEY
BOYS

$$\begin{aligned} u &= x-1 & v &= (x^2+7x)^5 \\ u' &= 1 & v' &= 5(2x+7)(x^2+7x)^4 \\ y' &= (x^2+7x)^5 + 5(2x+7)(x^2+7x)^4 \\ &= (x^2+7x)^4 (x^2+7x + 5(2x+7)) \\ &= (x^2+7x)^4 (x^2+7x+10x+7) \\ &= (x^2+7x)^4 (x^2+17x+7) \end{aligned}$$



Introduction to Calculus selective

Part 8: Quotient Rule

Unit

TEACHER'S COPY

Q1

Differentiate with respect to x . 2

ACE

$$\frac{x+1}{x-1}$$

$$\begin{aligned}\frac{d}{dx}\left(\frac{x+1}{x-1}\right) &= \frac{(x-1)1 - (x+1)1}{(x-1)^2} \\ &= \frac{x-1-x-1}{(x-1)^2} \\ &= \frac{-2}{(x-1)^2}\end{aligned}$$

Q2

Differentiate the following $y = \frac{x+1}{x-2}$

AMP

$$\begin{aligned}\frac{dy}{dx} &= \frac{(x-2) \cdot 1 - (x+1) \cdot 1}{(x-2)^2} \checkmark \\ &= \frac{x-2-x-1}{(x-2)^2} \\ &= \frac{-3}{(x-2)^2} \checkmark\end{aligned}$$

Q3

Differentiate: $\frac{2x}{1+x^2}$

(2)

ASCHAM

$$\begin{aligned}y' &= \frac{2(1+x^2) - 2x \cdot 2x}{(1+x^2)^2} \\ &= \frac{2-2x^2}{(1+x^2)^2}\end{aligned}$$

Q4

Find the derivative of $y = \frac{x+3}{x^2+1}$

2

ASCHAM

$$\begin{aligned}y &= \frac{x+3}{x^2+1} \\ y' &= \frac{(x^2+1) \cdot 1 - (x+3) \cdot 2x}{(x^2+1)^2} \\ &= \frac{x^2+1-2x^2-6x}{(x^2+1)^2} \\ &= \frac{-x^2-6x+1}{(x^2+1)^2}\end{aligned}$$



Introduction to Calculus selective

Part 8: Quotient Rule

Unit

TEACHER'S COPY

Q5

Differentiate $\frac{x^2}{1+x}$

(2)

ASCHAM

$$\begin{aligned} \frac{d}{dx} \frac{x^2}{1+x} &= \frac{(1+x)2x - x^2}{(1+x)^2} \checkmark \\ &= \frac{2x + 2x^2 - x^2}{(1+x)^2} \\ &= \frac{2x + x^2}{(1+x)^2} \checkmark \end{aligned} \quad (2)$$

Q6

Differentiate $y = \frac{5x^2}{2x+7}$

2

BAULKHAM

$$\begin{aligned} \frac{dy}{dx} &= \frac{10x(2x+7) - 2x \cdot 5x^2}{(2x+7)^2} \checkmark \\ &= \frac{20x^2 + 70x - 10x^2}{(2x+7)^2} \\ &= \frac{10x^2 + 70x}{(2x+7)^2} \checkmark \end{aligned}$$

Q7

Differentiate $y = \frac{x^2-7}{2x+3}$

2

BAULKHAM

$$\begin{aligned} y' &= \frac{2x(2x+3) - (x^2-7) \times 2}{(2x+3)^2} \checkmark \\ &= \frac{4x^2 + 6x - 2x^2 + 14}{(2x+3)^2} \\ &= \frac{2x^2 + 6x + 14}{(2x+3)^2} \checkmark \end{aligned}$$

Q8

Differentiate with respect to x : $\frac{x^2}{x-3}$

3

CRANBROOK

$$\begin{aligned} &\frac{vu' - uv'}{v^2} \\ &\frac{2x(x-3) - x^2}{(x-3)^2} \checkmark \\ &\frac{2x^2 - 6x - x^2}{(x-3)^2} \Rightarrow \frac{x^2 - 6x}{(x-3)^2} \checkmark \end{aligned}$$



Introduction to Calculus selective

Part 8: Quotient Rule

Unit

TEACHER'S COPY

Q 9

Find the derivative of: $\frac{2x-5}{3x+7}$

GIRAWEEEN

$$\begin{aligned} y &= \frac{2x-5}{3x+7} \\ y' &= \frac{v u' - u v'}{v^2} \\ &= \frac{2(3x+7) - 3(2x-5)}{(3x+7)^2} \\ &= \frac{6x+14-6x+15}{(3x+7)^2} \\ &= \frac{29}{(3x+7)^2} \quad (3) \end{aligned}$$

Q10

Differentiate: $y = \frac{4x^2-2}{x^2+5}$

GIRAWEEEN

$$\begin{aligned} \text{(iii)} \quad y &= \frac{4x^2-2}{x^2+5} \\ y' &= \frac{(x^2+5) 8x - (4x^2-2) 2x}{(x^2+5)^2} \\ &= \frac{8x^3+40x-8x^3+4x}{(x^2+5)^2} \\ &= \frac{44x}{(x^2+5)^2} \quad (3) \end{aligned}$$

Q11

Differentiate with respect to x : $\frac{3x+1}{x^2-1}$

INDEPENDENT

$$\begin{aligned} &= \frac{3(x^2-1) - 2x(3x+1)}{(x^2-1)^2} \\ &= \frac{3x^2-3-6x^2-2x}{(x^2-1)^2} \\ &= \frac{-3x^2-2x-3}{(x^2-1)^2} \end{aligned}$$

Q12

Find the derivative of $\frac{2x^2+1}{5-3x^2}$

2

NSBHS

$$\begin{aligned} &\frac{d}{dx} \left(\frac{2x^2+1}{5-3x^2} \right) \\ &= \frac{\left(\frac{d}{dx} 2x^2+1 \right) (5-3x^2) - \left(\frac{d}{dx} 5-3x^2 \right) (2x^2+1)}{(5-3x^2)^2} \\ &= \frac{(4x)(5-3x^2) - (-6x)(2x^2+1)}{(5-3x^2)^2} \\ &= \frac{20x-12x^3+12x^3+6x}{(5-3x^2)^2} \\ &= \frac{26x}{(5-3x^2)^2} \end{aligned}$$



Introduction to Calculus selective

Part 8: Quotient Rule

2 unit

TEACHER'S COPY

Q13

Differentiate with respect to x : $\frac{x-2}{4-x^3}$

2

NSGHS

$$\frac{(4-x^3) - (x-2)(-3x^2)}{(4-x^3)^2}$$

$$\frac{4-x^3+3x^3-6x^2}{(4-x^3)^2}$$

$$\frac{2x^3-6x^2+4}{(4-x^3)^2}$$

Q14

Differentiate $y = \frac{6x-1}{4-2x}$

2

SCEGGS

$$y = \frac{6x-1}{4-2x} \quad \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(4-2x) \times 6 - (6x-1) \times -2}{(4-2x)^2}$$

$$= \frac{24-12x+12x-2}{(4-2x)^2}$$

$$= \frac{22}{(4-2x)^2} \quad \checkmark$$

Q15

Differentiate $y = \frac{1-5x}{x+3}$

SYDNEY GRAMMAR

$$\frac{dy}{dx} = \frac{(x+3) \times (-5) - (1-5x)}{(x+3)^2} \quad \checkmark$$

$$= \frac{-16}{(x+3)^2} \quad \checkmark$$

Q16

Differentiate with respect to x .

$$\frac{4x+5}{2x^2+5x}$$

BARKER

$$y = \frac{4x+5}{2x^2+5x}$$

$$y' = \frac{(2x^2+5x) \times 4 - (4x+5)(4x+5)}{(2x^2+5x)^2}$$

$$= \frac{4(2x^2+5x) - (4x+5)^2}{(2x^2+5x)^2}$$



Introduction to Calculus selective

Part 8: Quotient Rule

Unit

TEACHER'S COPY

Q17

JAMES
RUSE

Find $f'(2)$ if $f(x) = \frac{2x-1}{x+1}$.

$$f(x) = \frac{2x-1}{x+1}$$

$$f'(x) = \frac{(x+1) \cdot 2 - (2x-1) \cdot 1}{(x+1)^2}$$

$$= \frac{3}{(x+1)^2}$$

$$\therefore f'(2) = \frac{1}{3}$$

Q18

HURLSTONE

Which of the following would represent $f'(x)$ if $f(x) = \frac{g(x)}{h(x)}$?

(A) $f'(x) = h(x) \times g'(x) + g(x) \times h'(x)$

(B) $f'(x) = \frac{h(x) \times g'(x) + g(x) \times h'(x)}{(h(x))^2}$

(C) $f'(x) = h(x) \times g'(x) - g(x) \times h'(x)$

(D) $f'(x) = \frac{h(x) \times g'(x) - g(x) \times h'(x)}{(h(x))^2}$

Q19

HURLSTONE

Differentiate: $\frac{4x^5 - 2}{1 - x}$

$$\begin{aligned} \frac{d}{dx} \frac{4x^5 - 2}{1 - x} &= \frac{(1-x) \cdot 20x^4 - (4x^5 - 2) \cdot (-1)}{(1-x)^2} \\ &= \frac{20x^4 - 20x^5 + 4x^5 - 2}{(1-x)^2} \\ &= \frac{20x^4 - 16x^5 - 2}{(1-x)^2} \end{aligned}$$

Q20

HURLSTONE

(i) Use the quotient rule to show that if $y = \frac{x}{(2x-3)^3}$, $\frac{dy}{dx} = \frac{-4x-3}{(2x-3)^4}$.

(ii) Hence, find the equation of the normal to the curve $y = \frac{x}{(2x-3)^3}$, at the point $(1, -1)$.

$$\begin{aligned} \text{(b)(i) } y &= \frac{x}{(2x-3)^3} \\ \frac{dy}{dx} &= \frac{(2x-3)^3 \times 1 - x \cdot 3(2x-3)^2 \cdot 2}{(2x-3)^6} \\ \frac{dy}{dx} &= \frac{(2x-3)^2 ((2x-3) - 6x)}{(2x-3)^6} \\ \therefore \frac{dy}{dx} &= \frac{-4x-3}{(2x-3)^4} \end{aligned}$$

(ii) Gradient of tangent at $x=1$ is $\frac{dy}{dx} = \frac{-4-3}{(2-3)^4} = -7$

$\therefore$ gradient of normal at $x=1$ is $\frac{1}{7}$.

$\therefore$ equation of normal is $y+1 = \frac{1}{7}(x-1)$

Equation is $x-7y-8=0$.



Introduction to Calculus selective

Part h: Rates of Change

Unit

TEACHER'S COPY

Q 1

Baulkham

Water was poured into a tank for 30 hours until it was full. At any time, t hours, the volume, V litres, of water in the tank was given by $V = 2(20t - t^2 + 100)$

- How much water was in the tank initially?
- At what rate was water poured into the tank at the end of 5 hours?

$$\therefore V = 2(20t - t^2 + 100)$$

$$i) t = 0$$

$$\therefore V = 2(0 + 100) = 200 \text{ l}$$

$$ii) V = 2(20t - t^2 + 100)$$

$$\frac{dV}{dt} = 2(20 - 2t)$$

$$\text{at } t = 5$$

$$\frac{dV}{dt} = 2(20 - 10) = 20 \text{ l/h}$$

Q 2

Caringbah

When a balloon is being filled with helium, its volume at time t is given by

$$V = \frac{\pi t^3}{12} \text{ cm}^3, \text{ where } t \text{ is in seconds.}$$

Find the rate at which the balloon is being filled when $t = 1$.

$$V = \frac{\pi t^3}{12} \text{ cm}^3 \rightarrow \frac{dV}{dt} = \frac{\pi t^2}{4}$$

$$\text{when } t = 1, \frac{dV}{dt} = \frac{\pi}{4} \text{ cm}^3/\text{s}$$

Q 3

Gosford

A rain water tank which is full is drained so that at time ' t ' minutes, the volume of water V in litres is given by

$$V = 500 \left(1 - \frac{t}{60}\right)^2 \text{ for } 0 \leq t \leq 60.$$

- How much water was initially in the tank?
- After how many minutes was the tank half full?
- At what rate was the water draining when the time is 58 min.

$$t = 0$$

$$i) V = 500$$

$$ii) V = 250 \text{ for half.}$$

$$250 = 500 \left(1 - \frac{t}{60}\right)^2$$

$$\frac{1}{2} = \left(1 - \frac{t}{60}\right)^2$$

$$\pm \sqrt{\frac{1}{2}} = 1 - \frac{t}{60}$$

$$\frac{t}{60} = 1 - \frac{1}{\sqrt{2}}$$

$$t = 0.29 \times 60$$

$$= 17.57 \text{ min}$$

$$iii) V = 500 \left(1 - \frac{t}{60}\right)^2$$

$$\frac{dV}{dt} = 1000 \left(1 - \frac{t}{60}\right) \times -\frac{1}{60}$$

$$= -\frac{1000}{60} \left(1 - \frac{t}{60}\right)$$

$$t = 58 \frac{dV}{dt} = -\frac{1000}{60} \left(1 - \frac{58}{60}\right)$$

$$= \frac{5}{9} \text{ Litres/min}$$

Q 4

PEM

Water flows into an empty container, so that after t minutes the volume V of water in litres is given by

$$V = \frac{12t^2}{t+4} \text{ for } t \geq 0.$$

What is the rate at which the water is flowing into the container after 1 minute?

(A) 4.32 L/min

(C) 21.6 L/min

(B) 4.32 min/L

(D) 108 L/min



Introduction to Calculus selective

Part h: Rates of Change

Unit

TEACHER'S COPY

Q 5

St
George

A large tank of liquid which contains L litres of a toxic chemical is being drained. The amount of chemical in the tank over time t minutes, is given by:

$$L = 110(20 - t)^2.$$

- i) At what rate is the water draining out of the tank after 5 minutes?
ii) How long will it take for the tank to be completely empty?

a(i)

$$L = 110(20 - t)^2$$

$$\therefore \text{Rate} = \frac{dL}{dt} = 110 \times 2(20 - t) \times -1$$

$$= -220(20 - t)$$

$$\therefore \text{at } t = 5$$

$$\frac{dL}{dt} = -220(20 - 5) = -3300 \text{ L / Min}$$

a(ii)

For empty tank: $L = 0$

$$\text{i.e. } 110(20 - t)^2 = 0$$

$$(20 - t)^2 = 0$$

$$\therefore t = 20 \text{ minutes}$$

Q 6

Sydney
Boys

The volume $V \text{ cm}^3$ of a balloon is increasing such that the volume at any time t seconds is given by

$$V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2}$$

Find the rate at which the volume is increasing when $t = 2$.

The volume $V \text{ cm}^3$ of a balloon is increasing such that the volume at any

time t seconds is given by

$$V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2}$$

Find the rate at which the volume is increasing when $t = 2$.

$$\frac{dV}{dt} = \pi t^2 - \frac{\pi t}{3}$$

$$t = 2 \Rightarrow \frac{dV}{dt} = \pi \times 2^2 - \frac{\pi \times 2}{3}$$

$$= \frac{10\pi}{3} \text{ cm}^3/\text{s}$$

Q 7

Sydney
Boys

The number of litres of water in a spa, t minutes after it has started to drain, is given by the equation $L = 60(14 - t)^2$.

- (i) Calculate the initial volume of water in the spa (leaving your answer in kilolitres). (1)
(ii) At what rate does the water drain out of the spa after 8 minutes? (2)

$$L = 60(14 - t)^2$$

$$\begin{aligned} \text{(i) when } t=0, L &= 60 \times 14^2 \\ &= 11760 \text{ L} \\ &= 11.76 \text{ kL. } \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \frac{dL}{dt} &= 60 \times 2(14 - t)^1 \times -1 \\ &= -120(14 - t) \end{aligned}$$

$$\text{when } t=8, \frac{dL}{dt} = -120 \times 6 = -720 \text{ L/min } \textcircled{2}$$



Introduction to Calculus selective

Part h: Rates of Change

Unit

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Q 8

Sydney
Grammar

A fuel tank ruptured and as a consequence the fuel began to leak out. The volume V litres of fuel remaining in the tank t minutes after the rupture was given by

$$V = 2000 - 100t - 10t^2.$$

- How much fuel was in the tank when the rupture occurred?
- What percentage of the fuel remained in the tank after 4 minutes?
- At what rate was the fuel leaking out after 4 minutes?
- How long did it take for the tank to empty?

(i) When $t=0$,
 $V = 2000$ L ✓

(ii) When $t=4$,
 $V = 2000 - 400 - 160$
 $= 1440$
 so percentage remaining is
 $\frac{1440}{2000} \times \frac{100}{1} \% = 72\%$. ✓

(iii) $\frac{dV}{dt} = -100 - 20t$
 When $t=4$,
 $\frac{dV}{dt} = -100 - 80$
 $= -180$
 So the fuel was leaking
 out at 180 L/min. ✓

(iv) When $V=0$,
 $0 = 2000 - 100t - 10t^2$
 $t^2 + 10t - 200 = 0$
 $(t+20)(t-10) = 0$ ✓
 $t \geq 0$ so $t=10$ ✓
 So the tank took 10 min to empty. ✓

Q 9

CEM

A tank is to be emptied by means of a control valve. The valve operates so that V litres, the volume of fluid remaining in the tank, varies with time t , measured in minutes, according to the relation

$$\frac{dV}{dt} = -kt, \text{ where } k \text{ is a constant.}$$

- Initially the tank contains 5000 litres of fluid. Show that after t minutes

$$V = 5000 - \frac{1}{2}kt^2$$

- If $k = 1.44$, at what rate will the tank be emptying when $V = 2000$?
- Find the time it takes to completely empty the tank? (Give your answer to the nearest minute).

[b][i] $V = 5000 - \frac{1}{2}kt^2$
 $\frac{dV}{dt} = -\frac{1}{2} \times 2kt$
 $= -kt \dots\dots\dots 1\text{m}$

[ii] $2000 = 5000 - 0.5 \times 1.44 \times t^2$
 $0.72t^2 = 5000 - 2000$
 $t^2 = \frac{3000}{0.72}$
 $t = 4166\frac{2}{3} \text{ mins} \dots\dots\dots 1\text{m}$
 $\frac{dV}{dt} = -1.44 \times 4166\frac{2}{3}$
 $= -6000 \text{ litres/min} \dots\dots\dots 1\text{m}$

[iii] When $V = 0$, $t^2 = \frac{5000 \times 2}{1.44}$
 $t = 6944.4 \text{ mins} \dots\dots\dots 1\text{m}$



Introduction to Calculus selective

Part h: Rates of change

2 unit

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Q 10

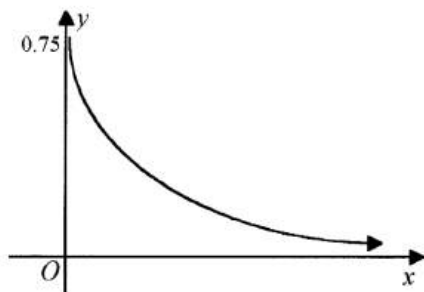
CSSA

The volume, V , of water in a dam at time t was monitored over a period of time. When monitoring began, the dam was 75% full. During the monitoring period, the volume decreased at an increasing rate due to a long period of drought.

- (i) What does this tell us about $\frac{dV}{dt}$ and $\frac{d^2V}{dt^2}$?
- (ii) Sketch the graph of V against t .

(b) (i) $\frac{dV}{dt} < 0$, $\frac{d^2V}{dt^2} > 0$

(ii)



Q 11

CSSA

The population (P) of a coastal town is increasing at a decreasing rate.

Comment on $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ for this function.

$\frac{dP}{dt} > 0$ because the function is increasing

$\frac{d^2P}{dt^2} < 0$ because the function is increasing at a *decreasing* rate

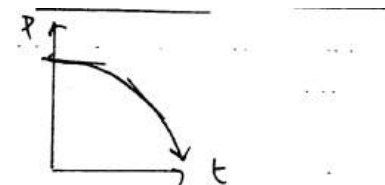
3

Q 12

CSSA

The population of a small country town is decreasing at an increasing rate.

Given that N is the population of the town at a given time t , what does this statement imply about $\frac{dN}{dt}$ and $\frac{d^2N}{dt^2}$?



$\frac{dN}{dt} < 0$ i.e. population is decreasing

$\frac{d^2N}{dt^2} < 0$

Q 13

CSSA

The price of one tonne of copper, $\$P$, was studied over a period of t years.

- (i) Throughout the period of study $\frac{dP}{dt} > 0$. What does this say about the price of copper?
- (ii) It was also observed that the rate of change of the price of copper decreased over the period of study.

What does this statement imply about $\frac{d^2P}{dt^2}$?

(7) (a) (i) The price is increasing.

(ii) $\frac{d^2P}{dt^2} < 0$



Introduction to Calculus selective

Part h: Rates of Change

Unit

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Q 14

CSSA

A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation:

$$L = 120(40 - t)^2$$

- At what rate is the water draining out of the pool when $t = 6$ minutes?
- How long will it take for the pool to be completely empty?

$$(i) \quad \frac{dL}{dt} = -240(40 - t)$$

at $t = 6$

$$\frac{dL}{dt} = -240(40 - (6))$$

$$\begin{aligned} &= -240(34) \\ &= -8160 \text{ L/min} \end{aligned}$$

- empty when $L = 0$

$$0 = 120(40 - t)^2$$

$$0 = (40 - t)$$

$$t = 40 \text{ mins}$$

Q 15

Geffernan

If water drains from a cylindrical tank according to the formula

$$V = 5000 \left(1 - \frac{t}{40}\right)^2, \text{ where } V \text{ is the volume of water in the tank}$$

at any time t . V is in litres and t in minutes.

- How much water is initially in the tank? 1
- How long will it take to empty the tank. 1
- Find the rate at which the water is flowing out of the tank after 10 minutes 2

$$(i) \text{ when } t = 0, V = 5000 \text{ L } \checkmark$$

$$(ii) \quad 0 = 5000 \left(1 - \frac{t}{40}\right)^2$$

$$\left(1 - \frac{t}{40}\right)^2 = 0$$

$$t = 40 \text{ min } \checkmark$$

$$\begin{aligned} (iii) \quad \frac{dV}{dt} &= 5000 \times 2 \left(1 - \frac{t}{40}\right) \times -\frac{1}{40} \\ &= -250 \left(1 - \frac{t}{40}\right) \checkmark \end{aligned}$$

$$\begin{aligned} \text{when } t = 10, \frac{dV}{dt} &= -187 \frac{1}{2} \\ &= 187 \frac{1}{2} \text{ L/min } \checkmark \end{aligned}$$



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Part h: Rates of change

2 unit

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Q 16

James
Ruse

The price of gold, \$P\$, was studied over a period of \$t\$ years.

- (i) Throughout the period of study $\frac{dP}{dt} > 0$. What does this say about the price of gold?
- (ii) It was also observed that the rate of change of the price of gold decreased over the period of study.
- What does this statement imply about $\frac{d^2P}{dt^2}$?

(a) The price of gold, \$P\$, was studied over a period of \$t\$ years.

- (i) Throughout the period of study $\frac{dP}{dt} > 0$. What does this say about the price of gold?

Price of gold is increasing

- (ii) It was also observed that the rate of change of the price of gold decreased over the period of study.

What does this statement imply about $\frac{d^2P}{dt^2}$?

$$\frac{d^2P}{dt^2} < 0$$

Q 17

PLC

In many situations involving environmental pollution much of the pollutant can be removed from the air or water for a reasonable cost, but the last part of the pollutant is very expensive to remove.

One cost function for this is given by $y = \frac{6x}{102-x}$, where \$y\$ is the cost in thousands of dollars of removing \$x\$ percent of a given pollutant.

- (i) Using this function find the cost of removing 90% of the pollutant. 1
- (ii) What percentage of pollutant can be removed for \$10 000? 2
- (iii) Sketch the cost function, clearly indicating its domain and range. (You may assume that it is a monotonic increasing function) 2

$$(a) y = \frac{6x}{102-x}$$

$$(i) x = 90, y = \frac{6 \times 90}{102-90} = 45$$

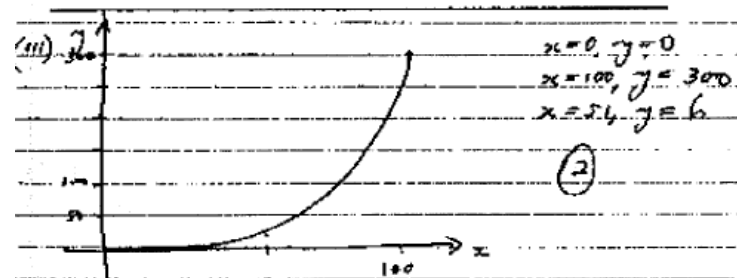
So cost = \$45 000

$$(ii) y = 10, 10 = \frac{6x}{102-x}$$

$$1020 - 10x = 6x$$

$$16x = 1020$$

$$x = 63.75\%$$





Introduction to Calculus selective

Part h: Rates of Change

Unit

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Q 18

PLC

The rate at which a reservoir is being filled is given by

$$V'(t) = 150t^{-\frac{1}{4}} + 10 \text{ litres/sec.}$$

- Find $V(t)$, the volume of water in the reservoir at time t secs, given that the reservoir holds 3 000 litres after 16 seconds.
- How much more will be added to the reservoir, to the nearest 100 litres, by the end of 15 minutes?

(i) $V'(t) = 150t^{-\frac{1}{4}} + 10$
 $V(t) = 150 \cdot \frac{4}{3} t^{\frac{3}{4}} + 10t + C$
 $V(16) = 3000$
 $3000 = 200(16)^{\frac{3}{4}} + 160 + C$
 $= 1600 + 160 + C \therefore C = 1240$
 $V(t) = 200t^{\frac{3}{4}} + 10t + 1240$

(ii) $15 \text{ mins} \Rightarrow t = 900$
 $V(900) = 200 \cdot 30\sqrt{30} + 9000 + 1240$
 $\therefore \Delta V(900) = V(16) = 6000 + 7240$
 $\approx 40100 \text{ L.}$

Q 19

Roseville

Moss, Kuswadi & Ryan Pty. Ltd. went into business on the first of January. The company profits increased each month until September (\$500, 000). Due to industrial strikes, their supply of goods was depleted and there was a sudden decline in profitability until December when they just managed to break even.

- Draw a possible graph showing profit against time (in months).
- Comment on the nature of $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ over time.

